WesChat 1.2 | Matrix Operations Cheat Sheet

Notation: L = sequence length; D = model width; H = heads; d_h = D/H; F = MLP width; V = vocabulary size. Matrix shapes omit batch dimension.

1. Forward propagation — one Transformer block

Operation	Matrix dimensions	Output
Input X	L × D	X
LayerNorm1(X)	L × D	X1
Q = X1 · Wq	(L × D)(D × D)	L × D
K = X1 · Wk	(L × D)(D × D)	L × D
V_a = X1 · Wv	(L × D)(D × D)	L × D
Split Q, K, V into H heads	L × D	H × L × d_h
RoPE(Qh, Kh)	Per head: L × d_h	Same
Scores = Qh · Kh^T	(L × d_h)(d_h × L)	H × L × L
Scale, causal mask, softmax, dropout	H × L × L	Scores2
X2h = Scores2 · Vh	(L × L)(L × d_h)	H × L × d_h
Concatenate heads	H × L × d_h	X2: L × D
X3 = X2 · W0	(L × D)(D × D)	L × D
Residual: X4 = X + X3	L × D	X4
LayerNorm2(X4)	L × D	X5
Hidden1 = X5 · W1 + b1	(L × D)(D × F)	L × F
Activation, MLP dropout	L × F	Hidden2
X6 = Hidden2 · W2 + b2	(L × F)(F × D)	L × D
Residual dropout	L × D	X6
Residual: X7 = X4 + X6	L × D	X7

Repeat for all nBlock Transformer blocks.

2. Output head and loss

Operation	Matrix dimensions	Output
Logits = X7 · E^T	(L × D)(D × V)	L × V
Softmax(Logits)	L × V	Probabilities
Cross-entropy loss	L × V	Scalar loss
dLogits = (P − Y) / L	L × V	TopGradient

E is the tied embedding matrix (V × D); Y is the one-hot target matrix. Division by L assumes mean loss over all sequence positions.

WesChat 1.2 | Backpropagation

3. Reverse through the output head and one Transformer block. Accumulate gradients at both residual connections.

Backward operation	Matrix dimensions	Result
OUTPUT HEAD		
$dX7 = dLogits \cdot E$	$(L \times V)(V \times D)$	$L \times D$
$dE \mathrel{+}= dLogits^T \cdot X7$	$(V \times L)(L \times D)$	$V \times D$
MLP BACKWARD		
Residual split dX7	$L \times D$	dX4, dX6
Residual dropout backward	$L \times D$	dX6
$dW2 = Hidden2^T \cdot dX6$	$(F \times L)(L \times D)$	$F \times D$
db2 = row sum(dX6)	$L \times D$	D
$dHidden2 = dX6 \cdot W2^T$	$(L \times D)(D \times F)$	$L \times F$
MLP dropout + activation backward	$L \times F$	dHidden1
$dW1 = X5^T \cdot dHidden1$	$(D \times L)(L \times F)$	$D \times F$
db1 = row sum(dHidden1)	$L \times F$	F
$dX5 = dHidden1 \cdot W1^T$	$(L \times F)(F \times D)$	$L \times D$
LayerNorm2 backward	$L \times D$	Accumulate into dX4
ATTENTION BACKWARD		
Residual split dX4	$L \times D$	dX, dX3
$dW0 = X2^T \cdot dX3$	$(D \times L)(L \times D)$	$D \times D$
$dX2 = dX3 \cdot W0^T$	$(L \times D)(D \times D)$	$L \times D$
Split dX2 into heads	$L \times D$	$H \times L \times d_h$
$dScores2 = dX2h \cdot Vh^T$	$(L \times d_h)(d_h \times L)$	$H \times L \times L$
$dVh = Scores2^T \cdot dX2h$	$(L \times L)(L \times d_h)$	$H \times L \times d_h$
Dropout, softmax, scale backward	$H \times L \times L$	dScores1
$dQh = dScores1 \cdot Kh$	$(L \times L)(L \times d_h)$	$H \times L \times d_h$
$dKh = dScores1^T \cdot Qh$	$(L \times L)(L \times d_h)$	$H \times L \times d_h$
Reverse RoPE; combine heads	$H \times L \times d_h$	dQ, dK, dV_a: $L \times D$
$dWq = X1^T \cdot dQ$	$(D \times L)(L \times D)$	$D \times D$
$dWk = X1^T \cdot dK$	$(D \times L)(L \times D)$	$D \times D$
$dWv = X1^T \cdot dV_a$	$(D \times L)(L \times D)$	$D \times D$
$dX1 = dQ \cdot Wq^T + dK \cdot Wk^T + dV_a \cdot Wv^T$	Three $L \times D$ products	$L \times D$
LayerNorm1 backward	$L \times D$	Accumulate into dX

LayerNorm backward also produces gamma/beta gradients (each length D). Backward RoPE reverses the rotation; attention-score gradients include scaling and masking.